

Cyber Security Internship – Task 5

Malware Types & Behaviour Analysis

1. Introduction

Malware is malicious software designed to damage systems, steal data, or disrupt normal operations. Understanding malware types and behavior helps in detecting and preventing cyber-attacks.

2. Types of Malware

Virus

A virus attaches itself to legitimate files and spreads when the file is executed.

Worm

A worm spreads automatically across networks without user interaction.

Trojan

A trojan disguises itself as legitimate software but performs malicious actions in the background.

Ransomware

Ransomware encrypts files and demands payment to restore access.

3. Malware Analysis Using VirusTotal

VirusTotal is an online service that analyzes files and hashes using multiple antivirus engines. A known malware hash was analyzed to understand detection and behavior.

4. Detection Report

The analyzed hash was detected as malicious by multiple antivirus engines. Different engines classified it as ransomware.

Detection Ratio: High (multiple vendors flagged it as malicious)

5. Behavior Indicators

Observed behavior includes:

- File encryption
- Network communication with external servers
- System modification

These behaviors are common in ransomware attacks.

6. Malware Lifecycle

1. Infection
2. Execution
3. Propagation
4. Damage
5. Persistence

7. Malware Spread Methods

- Email attachments
- Malicious links
- Infected downloads
- Network vulnerabilities

8. Prevention Methods

- Use antivirus software
- Keep systems updated
- Avoid unknown downloads
- Enable firewalls
- Regular backups

9. Conclusion

This task improved my understanding of malware types, detection techniques, and prevention methods using VirusTotal.